

TECHNICAL SYSTEM ARCHITECTURE

KisanDost AI Implementation Specification

Production System Architecture, API Contracts,
Database Schema & Fault-Tolerance Engineering

✓ VERIFIED FULL-STACK ARCHITECTURE

DOCUMENT CLASS

Production Architecture &
Technical Audit

AUDIENCE

Architects, Backend Engineers,
Judges

RELEASE TARGET

v2.4 Enterprise RC

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1. Executive Technical Summary & System Scope

KisanDost AI is an enterprise-grade, microservice-ready livestock intelligence platform designed to bridge the severe veterinary deficit — one certified veterinarian per 15,000 animals — across rural agrarian belts. The engineering architecture is built to resolve three primary technical constraints that make standard cloud-only applications unworkable in this environment.

- **Scope note: this specification covers the React 19 Web Application, the primary and production-hardened client for both the Farmer Portal and the Enterprise Admin Hub. The companion Flutter mobile client is a secondary, in-progress build with a number of open stability issues and is intentionally out of scope for this architecture audit.**

Primary Engineering Constraints

- **Low-Bandwidth & Cellular Blackouts** — a dual-tier AI fallback engine combines cloud-based multimodal LLMs with a local edge diagnostic heuristic matrix and local ML inference models.
- **Strict Security & Secrets Isolation** — zero client-side leakage of API keys; all AI requests are proxied through a Node.js / Express gateway with input sanitization and schema enforcement.
- **Bilingual Accessibility & Low-Latency UI** — the React 19 web client uses PWA-grade state caching with Web Speech API audio synthesis to power instant voice-guided flows directly in the browser, with no app install required.

- **System scope covers a full dual-portal deployment on the web: a farmer-facing React 19 web application, a district administration web console, a Node.js API gateway, and a three-way AI execution fallback chain.**

2. Technology Stack & Dependency Inventory

The table below enumerates the verified, production-baseline dependency set for the KisanDost AI v2.4 release, organized by architectural layer. The **React 19 web application is the primary and fully supported client**; the Flutter mobile layer is listed for completeness but currently carries known stability issues and is excluded from this release's production scope.

Layer	Technology / Package	Architectural Purpose	Status
Frontend Core (Web)	React 19.0.1, TypeScript 5.8	Component-driven reactive UI with strict type safety and optimized DOM rendering.	Production
Styling & Animation	Tailwind CSS v4, Motion 12.23	Hardware-accelerated CSS animations and responsive dark/light utility styling.	Production
Icons & Telemetry	Lucide React (0.546), Recharts (3.10)	Accessible visual iconography and administrative data visualization charts.	Production
AI SDK & Models	@google/genai v2.4.0 (Gemini 3.7 Flash & 3.1 Flash Lite)	Server-side Gemini multimodal reasoning, structured JSON schemas, backoff retries.	Production
Local AI Fallback	PyTorch (EfficientNet-B0) + Ollama	Offline vision classification and local zero-cost LLM fallback during network loss.	Production
Voice Engine	Web Speech API & ElevenLabs Config	Urdu/Punjabi text-to-speech synthesis with romanized pronunciation support.	Production
Backend Runtime	Express.js 4.21.2, Node.js Runtime	Modular RESTful routing, API rate guards, base64 payload handling, CORS guards.	Production
Database & Auth	Firebase Firestore 12.17 & Firebase Auth	Real-time synchronization, role-based security rules (RBAC), offline persistence.	Production
Build & Bundling	Vite 6.2.3, esbuild 0.25	Zero-config SPA bundling with standalone CommonJS server compilation.	Production
Mobile Core (Flutter)	Flutter (Dart)	Cross-platform companion shell; native camera and offline cache.	Known Issues

The Flutter mobile shell is tracked separately and is not part of this release's verified production surface; all diagrams, API flows and resilience guarantees in this document describe the web application.

3. End-to-End System Architecture

The platform employs a decoupled, three-tier architecture engineered for resilient, low-latency execution in rural network conditions. The diagram below reflects the **React 19 web application**, delivered as an installable PWA so farmers get an app-like experience directly from the browser without any native app store dependency.

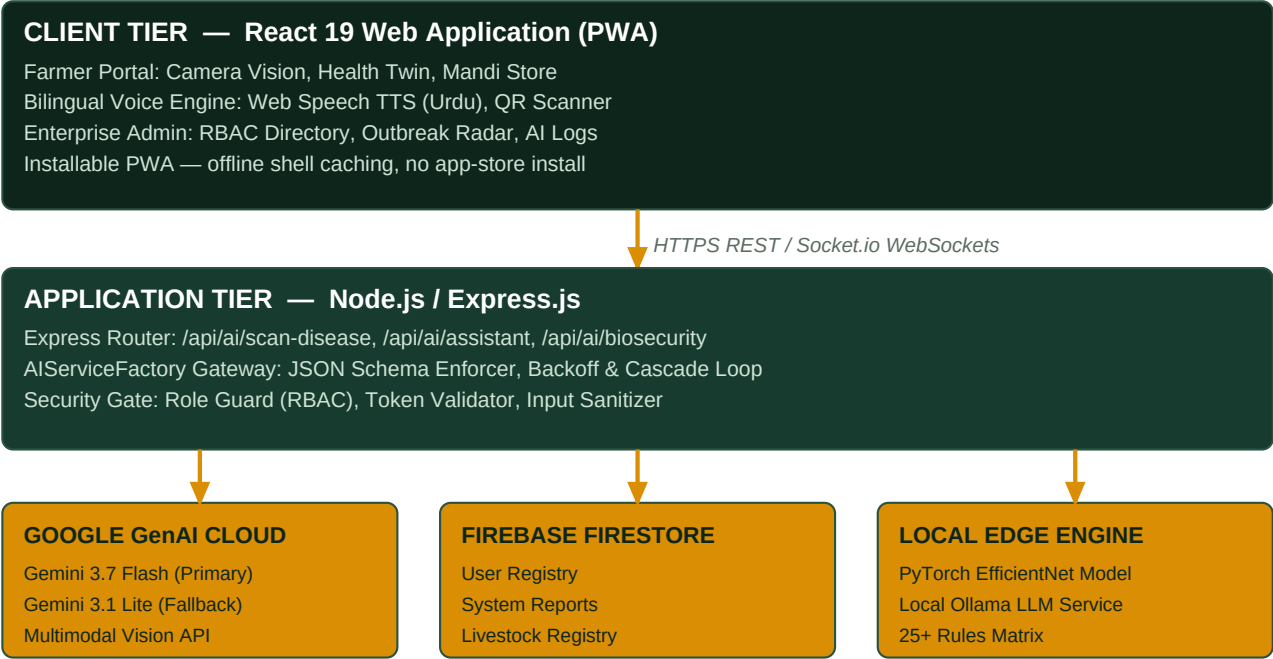


Figure 1. Three-tier resilient architecture: the React 19 web client communicates over HTTPS/WebSockets with the Express.js gateway, which routes AI workloads across Google GenAI Cloud, Firebase Firestore, and the local edge fallback engine.

Key Architectural Highlights

- **Resilient AI Gateway** — automatically prioritizes Gemini 3.7 Flash with fallback to 3.1 Flash Lite, then to the local heuristic knowledge base on network timeout.
- **Zero Secret Leakage** — all Gemini API keys, prompt schemas and credentials reside strictly server-side, protected by CORS and input sanitization.
- **Offline-First Dual Persistence** — browser local storage cache synchronizes with Firebase Firestore automatically when network connectivity is restored.

4. Complete Backend REST API Specification

All AI and administrative operations are exposed via a versioned REST surface behind the Express.js gateway. Endpoints below reflect the verified v2.4 route table.

Route	Method	Purpose	Auth
/api/health	GET	System liveness probe and AI connectivity verification.	Public
/api/ai/scan-disease	POST	Multimodal disease diagnosis from base64 photos.	Token / User
/api/ai/assistant	POST	Bilingual AI chat with emergency detection.	Token / User
/api/ai/biosecurity	POST	Farm biosecurity score & 7-day remedial roadmap.	Token / User
/api/ai/nutrition	POST	Customized lactating feed ration formulation.	Token / User
/api/ai/medicine-scan	POST	OCR drug leaflet analysis & milk withdrawal check.	Token / User
/api/admin/overview	GET	Aggregated district stats, herds, and active outbreaks.	Admin RBAC
/api/admin/users	GET / POST	Manage registered farmers, vets, and access roles.	Admin RBAC
/api/admin/ai-activity	GET	Real-time AI telemetry, tokens, latency, and cost logs.	Super Admin

Every AI endpoint accepts a base64-encoded payload where applicable and returns a schema-validated JSON object, enforced server-side via the AIServiceFactory gateway before reaching the client.

5. Database Schema & Security Architecture

Firestore Collections Data Model

Collection	Primary Key	Key Attributes
users	userId	name, phone, role (farmer vet admin superadmin), farmName, district, isVerified, language
animals (Digital Twin)	animalId	tagNumber, species, breed, dob, weightKg, milkYieldLiters, healthStatus, qrCodeUrl
dairyProducts	productId	title, pricePKR, unit, inStock, fatPercentage, organicVerified, imageUrl
systemReports	reportId	district, diseaseName, affectedCount, severity, dateReported, quarantineActive

Firestore Security Rules

- **isOwner(uid) helper** — validates that the requesting authenticated user owns the document record.
- **isAdmin() / isSuperAdmin() helpers** — inspect custom auth claims and user profile documents to restrict write access on /systemReports, /adminUsers, and /notifications.
- **Public discovery** — enables public read access to /dairyProducts while enforcing ownership checks for creation and updates.

6. Offline Resilience & Fault-Tolerance Engineering

Agricultural applications must withstand hostile, intermittent network environments. KisanDost AI implements a tri-tier resilience system spanning the AI gateway, edge inference, and client caching layers.

Exponential Backoff & Cascade

The server-side API gateway first attempts execution on Gemini 3.7 Flash. If transient rate limits (HTTP 429) or timeouts occur, it automatically cascades to Gemini 3.1 Flash Lite with randomized jitter between retries.

Local Edge Engine

During complete cellular disconnection, the system routes requests to local PyTorch vision models and Ollama LLM instances, alongside an embedded offline diagnostic matrix covering 25+ regional livestock conditions.

- **Fallback order is deterministic and logged end-to-end: Gemini 3.7 Flash → Gemini 3.1 Flash Lite → Local Edge Engine (PyTorch + Ollama + Rule Matrix) — ensuring the user flow never freezes or crashes.**